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Week 14 - Full Results Analysis

Interpretation of the final reconstructed cohort and robustness checks

1. Cohort

The analysis starts with 2,657 sprint rows. Thirty FUTURE rows, 14 ACTIVE rows, and three CLOSED rows with impossible timestamps are removed before label construction. One first eligible sprint per project is removed because there is no prior history. The final model frame has 2,602 sprints.

Step	Count
Raw sprint rows	2,657
Eligible closed sprints	2,610
Cold starts removed	8
Modeled sprints	2,602
Reconstructed commitments	20,603
Modeled zero-scope sprints	911
Modeled nonempty sprints	1,691

2. Main results

Late closure remains a moderate prediction problem. The main random-forest AUC is .705. Conditional spillover is easier to rank, but the transparent scope-only logistic model reaches .885 AUC, compared with .852 for random forest. The result favors parsimony: initial scope explains most of the measurable spillover signal.

Outcome / model	Test n	Positives	F1	ROC-AUC
Late closure - Random Forest	786	246	.562	.705
Late closure - XGBoost	786	246	.535	.695
Nonempty spillover - Always positive	511	482	.971	.500
Nonempty spillover - Scope-only logistic	511	482	.696	.885
Nonempty spillover - Random Forest	511	482	.875	.852

3. Uncertainty

Project-cluster intervals are wide enough that small model differences should not be overinterpreted. Late-closure random forest has a 95% AUC interval of .604-.783. Conditional spillover has .820-.922 for scope-only logistic regression and .797-.914 for random forest.

4. Temporal and cross-project stability

Forward cut points give late-closure RF AUCs from .669 to .711 and conditional-spillover RF AUCs from .846 to .856. Held-out-project performance varies much more, especially for late closure. That's an argument for validating locally, project by project — not for leaning on one tidy global number.

5. Provenance sensitivity

When no change history is on record, the static fallback is a guess — an assumption about identification, not something we can actually verify. Restrict to sprints where every committed item has observed Sprint history, and the test set shrinks to 356 sprints; AUC lands at .875 for the scope-only model, .850 for random forest. Push further and require both Sprint and Story Points history, and only 110 test sprints are left. AUC drops to .772 and .652. That's not noise — it's a real difference.

Subset	Test n	Scope LR AUC	RF AUC
Main nonempty cohort	511	.885	.852
Observed Sprint history	356	.875	.850
Observed Sprint + Story Points history	110	.772	.652

6. Feature evidence

Committed issue count dominates conditional spillover by a wide margin — shuffle it, and AUC drops by .220 on average. Late closure runs on something else entirely: project identity and sprint length are what actually move the needle there.

7. Interpretation

No sophisticated model has cracked Scrum success here — that's not the finding. What the data actually shows is that reconstructed sprint-start scope already carries real signal about unfinished work, and a plain scope model captures most of it on its own. Late closure is a weaker, messier, more project-specific story by comparison. And because everything here is retrospective, none of it should be read as proof that handing these scores to Scrum teams would actually improve delivery.